TECHNICAL + MANAGEMENT ORALS

NASA Advanced Computing

Scientific service with visible accountability

A fictional proposal portfolio by Meghan Fasano

We connect NASA’s scientific mission to qualified people, dependable computing and visible accountability.

MEET THE PRESENTERS

Your team in the room

Four leaders.
One accountable team.

Maya Bennett

Program Manager

Accountability, staffing and the first 30 days

Daniel Reyes

NAS Site Delivery Lead

Local continuity and scientific support

Priya Shah

Technical Integration Lead

Application engineering and evidence for change

Jordan Ellis

NCCS Site Delivery Lead

Climate workflows and researcher experience

The service story

The mission connects every chapter.

01 **Mission + service**

02 People + delivery

03 Scientific services

04 Operations + assurance

05 Past performance

06 Key personnel

07 Government questions

NASA MISSION CONTEXT

Computing for NASA science

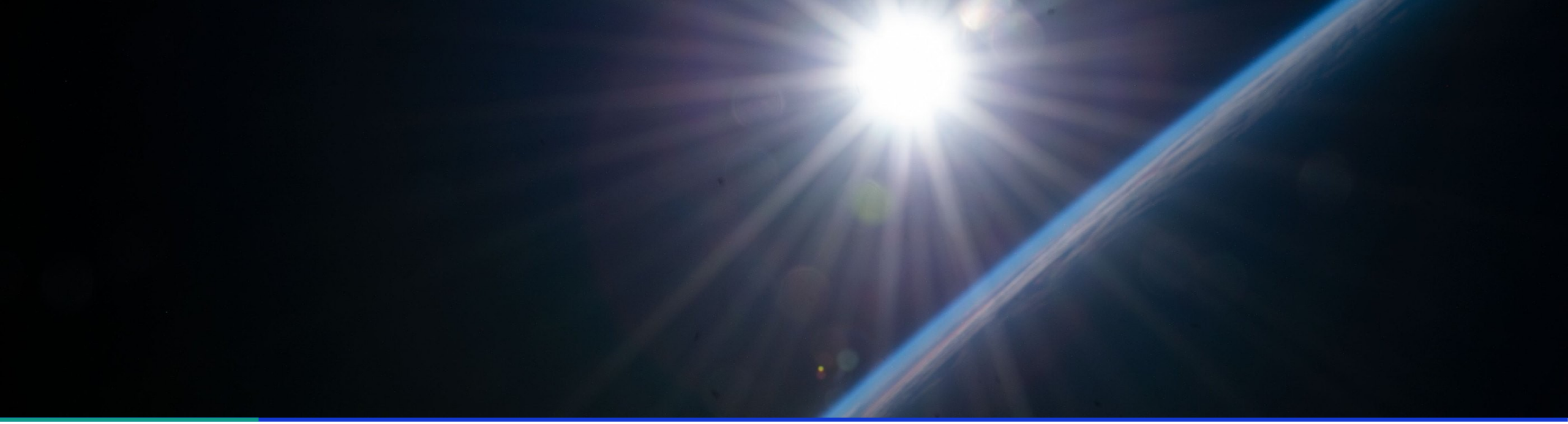
A clear path to validated scientific results.

Earth + climate

Reproducible models.
Research-ready outputs.

Aeronautics + exploration

Measured simulations.
Expert support for controlled change.



SECTION 01

Mission + service

The research objective stays visible through delivery.

- Scientific results
- Apps + workflows
- Data + workflows
- Compute + storage

- We carry the research objective through every service layer, from computing resources to scientific results.

An accountable science service

Computing capability, expert support and reproducible scientific results.

Capability

Compute, storage and networks

Expertise

Applications and scientific support

Science

Validated results and usable handoffs

- Computing capability becomes useful science when expert support connects resources to validated results.

Two centers. Local accountability

Shared expertise follows competence, access and capacity checks.

NAS Pleiades storage · NASA Ames/Dominic Hart

NAS / Ames

Local operations and scientific support
Control room: 24/7

Shared expertise

Common practices
Qualified specialists

Discover supercomputer · NASA Goddard

NCCS / Goddard

Local operations and scientific support
Help desk: 08:00–18:00 ET, Mon–Fri

- NASA receives local accountability at both centers, supported by qualified expertise shared across the team.

The service around the computer

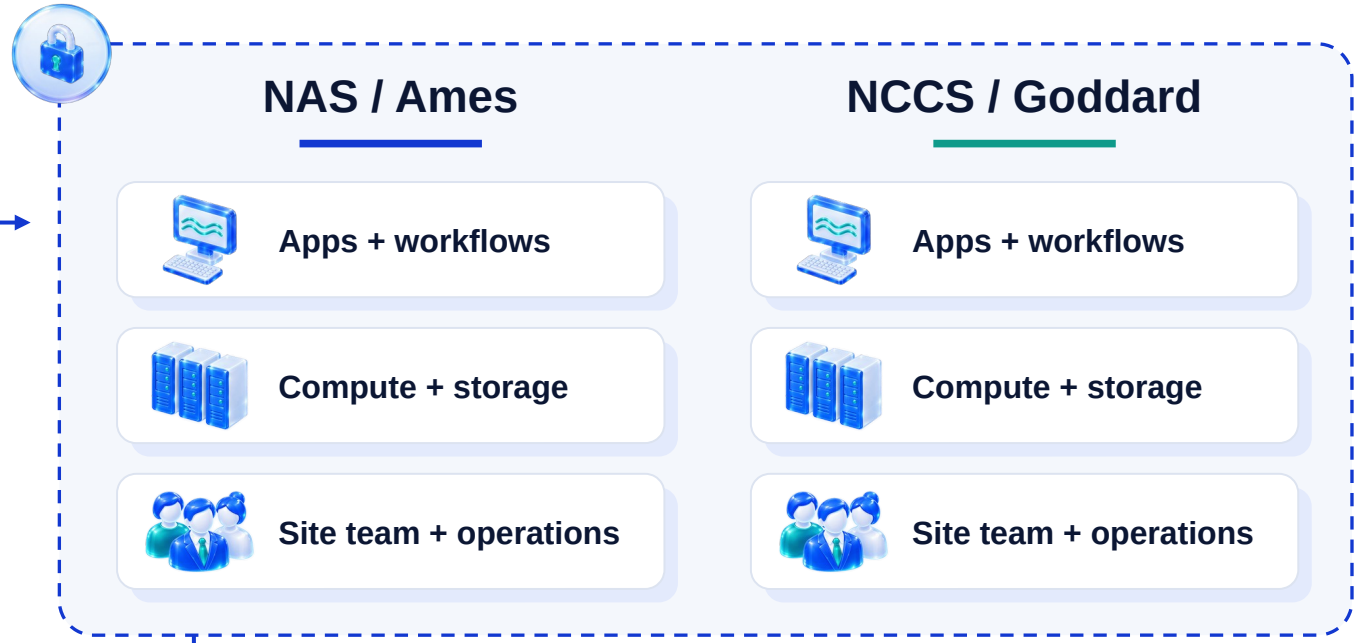
Local ownership at both centers. Security spans every service layer.

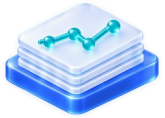
Scientific need

Researcher objective

Authorized access

Intake, accounts and allocation





Scientific output

Validation, provenance and reproducible handoff

- We support the whole research workflow, with authorized access and clear ownership around the computer.

SECTION 02

People + delivery

Qualified coverage, clear authority
and a controlled transition.

- Qualified people, clear authority and a controlled transition establish dependable delivery from the start.

One management chain

Each employee has one supervisor. Each task has one accountable lead.

Corporate support

HR, recruiting, legal, safety and specialist engineering

Maya Bennett

Program Manager
Overall contractor performance

Daniel Reyes

NAS Site Delivery Lead
Ames operations + support

Priya Shah

Technical Integration Lead
Cross-site dependencies

Jordan Ellis

NCCS Site Delivery Lead
Goddard operations + support

- Maya owns overall performance; Daniel, Priya and Jordan connect local delivery and technical integration to her.

Authorization and delivery

The Contracting Officer (CO) issues task orders. Government accepts completed work.

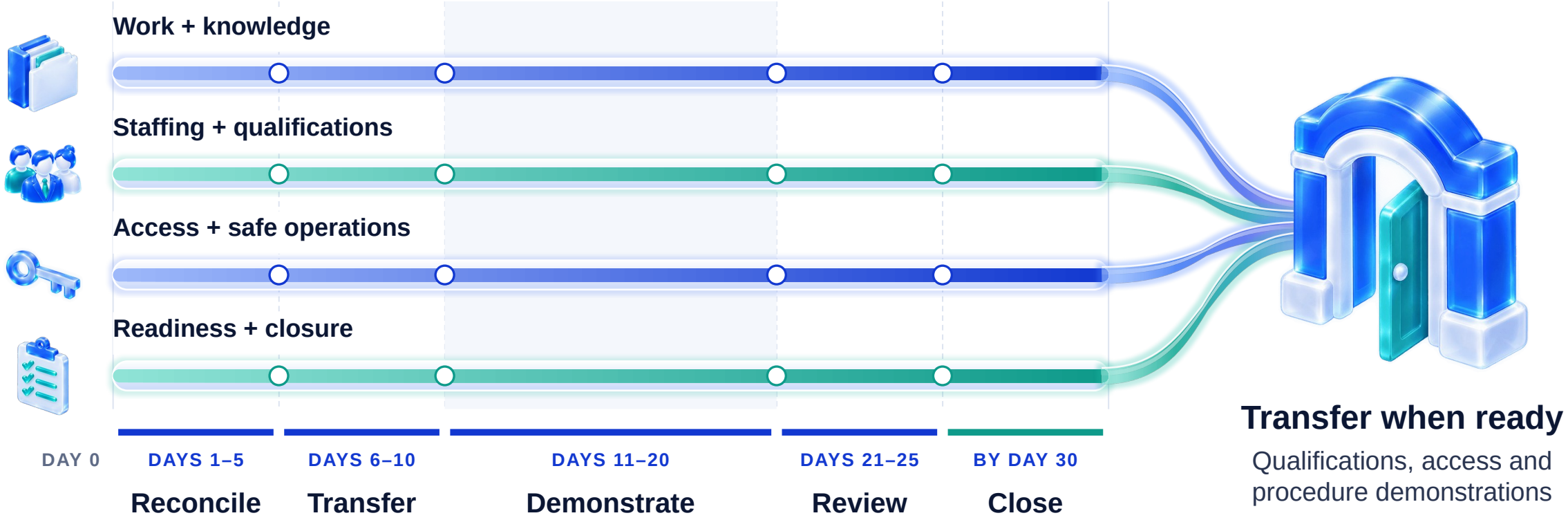
1	2	3	4	5	6	7
Request	**Plan**	**Authorize**	**Acknowledge**	**Execute**	**Verify**	**Accept**
Government need	Within 15 calendar days	CO issues or amends order	Within 5 calendar days of receipt	Authorized scope	Tests and defect closure	Government decision

Independent QA supports acceptance. Scope changes return to the CO.

- We turn authorized work into an accountable delivery record that ends with Government acceptance.

A controlled 30-day phase-in

Plan due at award. Day 0 is the contract effective date.



Transfer when ready

Qualifications, access and procedure demonstrations

- Responsibility transfers when people, access and procedures demonstrate readiness to protect ongoing research.

Qualified coverage before assignment

Operations, applications and user support each have a primary and alternate.

Primary + alternate

Named coverage for each function

Competence
Demonstrated skill

Access
Authorized entry

Availability
Capacity to deliver

Critical-role gap: same-day Program Manager review.

- Every assignment depends on demonstrated competence, approved access and an available qualified alternate.

PEOPLE + DELIVERY

Recruitment and incumbent continuity

A qualified pool becomes an accepted assignment through demonstrated readiness.

01 Build the qualified pool

Days 1–3: confirm gaps and open outreach. Use professional networks, universities and referrals.

02 Validate role readiness

By day 10: compare candidates with a common role exercise, structured interview and access requirements.

03 Confirm and onboard

Day 20: target selection. Day 30: target readiness. Qualified alternates protect essential work if acceptance falls short.

Illustrative assumption: 70% incumbent acceptance. Invented analogs: 32/46 and 13/18, or 70.3% combined. Actual decisions drive staffing.

- We turn a candidate pool into ready coverage through role evidence, selection and controlled onboarding.

Prime accountability with specialist depth

Orbital Method / 10%

Application porting and optimization

The prime holds all four key positions. Work packages define scope, interfaces, qualifications, access and corrective obligations.

Vector Ridge / 4%

Bounded research and emerging-computing experiments

Lumen Scientific / 6%

Visualization and scientific communication

The prime coordinates delivery and reviews actual effort and quality. Shares are illustrative labor effort, not dollar goals or certifications.

- Specialists add depth while the prime retains the key positions, delivery interfaces and responsibility for quality.

Knowledge continuity

An alternate demonstrates readiness before responsibility transfers.

Shared knowledge

01 Notice
Workload concentration and coverage risk

02 Develop
Mentoring and reviewed assignments

03 Demonstrate
Runbook exercise and reviewed work product

04 Cover
Qualified alternate with access and capacity

Any gap returns to development before the next transfer.

- Knowledge becomes continuity when an alternate can demonstrate the work and take responsibility for it.

SECTION 03

Scientific services

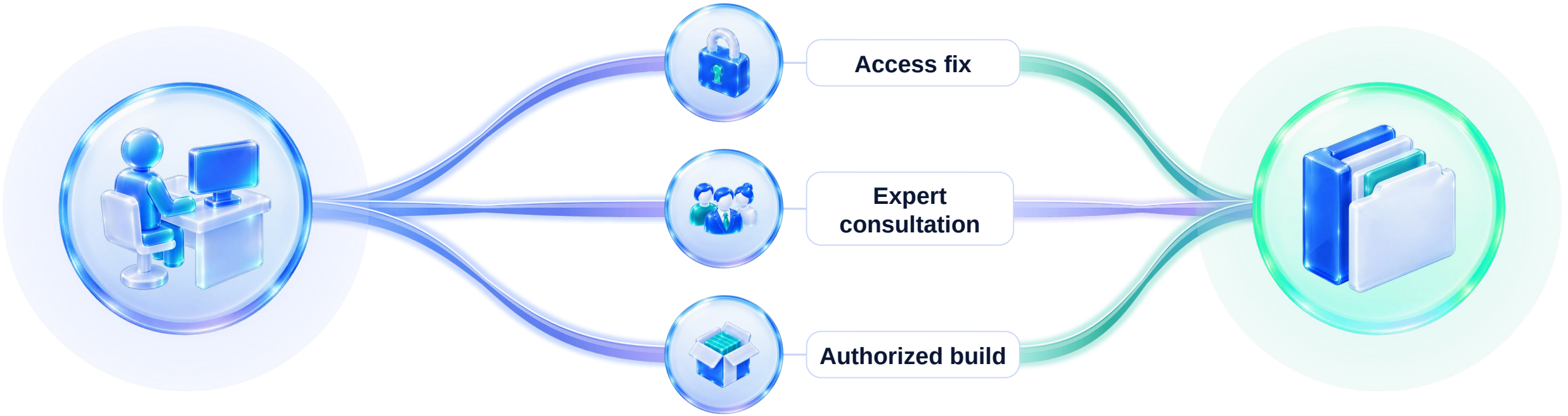
A clear path from research need
to validated delivery.

- **Validated results**
- **Scientific applications**
- **Data + provenance**
- **Authorized resources**

- Scientific services connect an authorized research need to an application, a traceable workflow and a valid result.

A path for every research request

One owner remains accountable through every handoff.



Scientific need

Objective and access triage

Reproducible result

Validation, provenance and reproduction instructions

- Each research request follows the right path, with one owner accountable through the final handoff.

Scientific data becomes an interpretable result

The investigator owns scientific meaning. Specialists preserve context through the complete workflow.

01 Prepare trustworthy data

Preserve versions, units, provenance and uncertainty. Support access and analysis of large research datasets.

02 Explore with the scientist

Use scientific views and hyperwall sessions to reveal behavior. The investigator validates the interpretation.

03 Deliver a reproducible result

Deliver versioned outputs, source references and usage guidance. Confirm release permissions and a support owner.

SOW 4.10–4.13 / Research data services are distinct from producing official NASA data products.

- We preserve data context so the investigator can interpret the result and another researcher can reproduce it.

Technology selected for NASA science

Candidate windows follow scientific constraints and reproducible baselines.

NASA science sets the clock

Years 2–10 depend on annual options, funding and authorization.

- NASA’s workloads and scientific constraints determine when a technology deserves further investment.

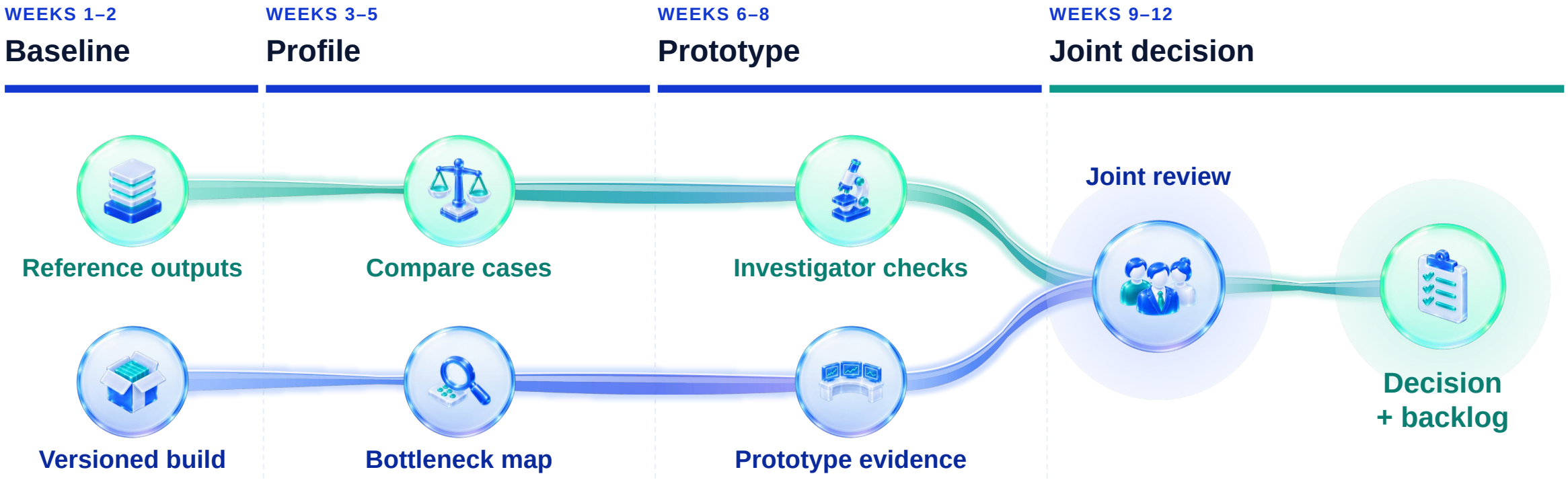
Evidence gates for adoption

Quarterly recommendations inform NASA authorization.

- A technology advances only when evidence supports scientific validity, operability and NASA authorization.

A 12-week application diagnostic

Starts after authorization and access to code, data and investigators.

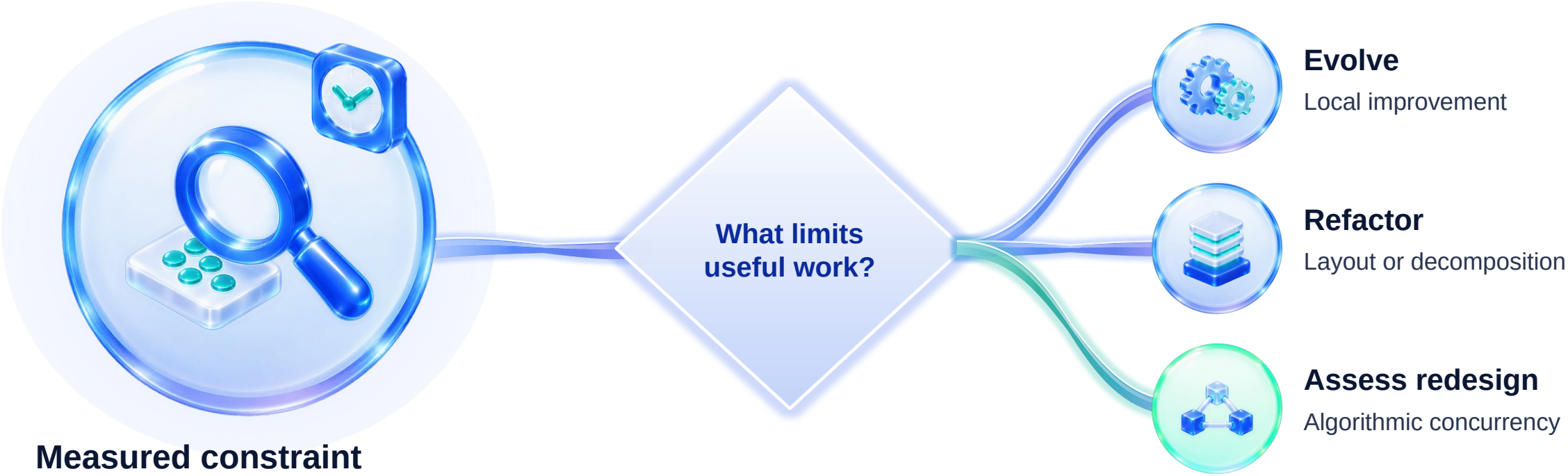


Scientific reference and engineering measurement advance together.

- A shared baseline and paired science checks turn application diagnosis into an informed engineering decision.

Engineering effort follows the bottleneck

Profiles and timing traces identify what limits useful work.



Measured constraint

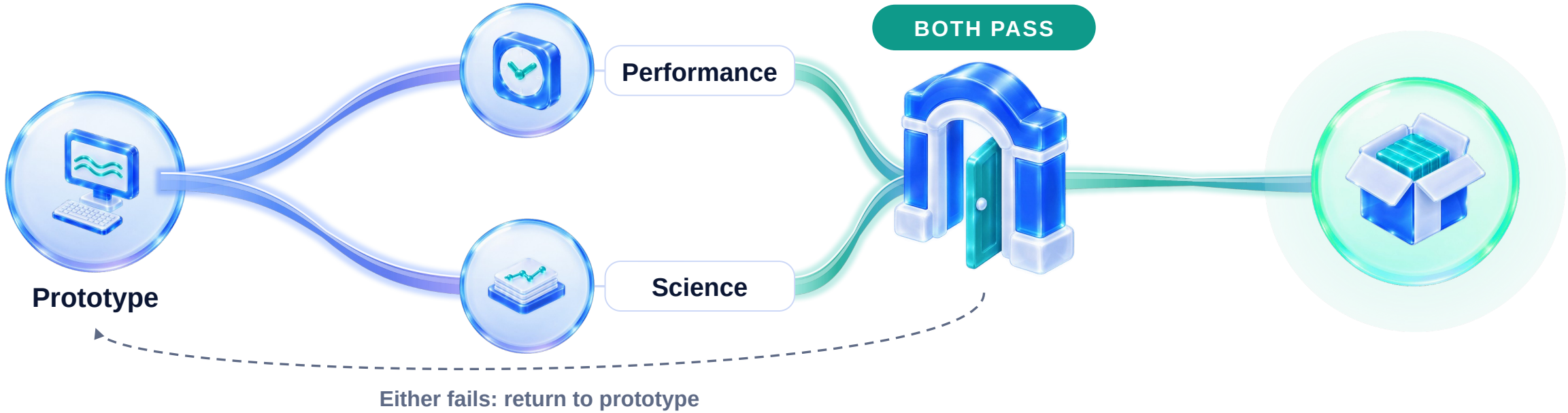
Compute, wait and I/O
Profiles + timing traces

Qualitative decision framework. No numerical performance profile is implied.

- Measured constraints guide the smallest effective engineering change while the scientific reference stays intact.

Performance and science must both pass

Acceptance requires useful speed and investigator-approved scientific validity.



01 Performance

Uninstrumented timing
Controlled restart

02 Science

Preserved reference
Approved tolerances

03 Handoff

Maintainable increment · build instructions
Training and rollback

- A faster application is ready for handoff only when useful performance and scientific validity both pass.

Exascale readiness depends on more than the processor

A recommendation carries its assumptions, remaining uncertainty and next test.

Validated recommendation

Science reference

Agreed cases, tolerances and investigator availability.

Software ecosystem

Versioned compilers, libraries, portability and regression tests.

Representative platform

Authorized access and allocations. Test at the intended scale.

Resilience + ownership

Checkpoint tests, trained support and a documented rollback.

RFP L.8 B2 / Missing code, scale or science evidence narrows the recommendation. Candidate innovations remain experiments until validated.

- Exascale readiness depends on the science, software, platform and recovery plan working together.

SECTION 04

Operations + assurance

Continuity, controlled change and inspectable decisions.

Records
Traceable work

Protection
Approved access

Control
Authorized change

- Records, protection and control make daily delivery visible and keep change accountable to NASA.

Staged service coordination

Common practices expand after demonstrated readiness. Local coverage stays protected.

DAY 30
Service baseline

DAY 60
Handoff pilot

DAY 90
Backup qualification

DAY 120
Deployment review

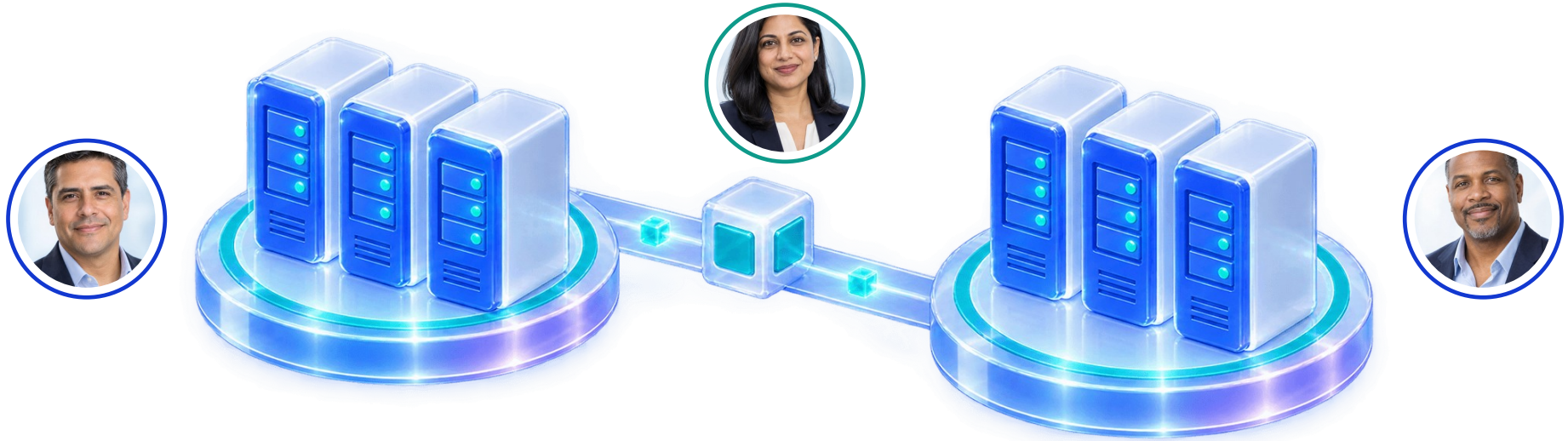
DAY 180
Handbook + benefits review

Failed qualification retains the current owner. This plan excludes system and application performance.

- Shared practices expand in stages as readiness is demonstrated and local service remains protected.

Cross-site improvement with protected local service

Daniel and Jordan retain local coverage. Priya coordinates a bounded pilot and an accepted handoff.



Local mission needs

Keep site-approved exceptions, access controls and staffing. Shared practices do not imply automatic workload failover.

Controlled handoff

Name a receiving owner and qualified alternate. Sequence one practice at a time with a reversible change.

Measured service benefit

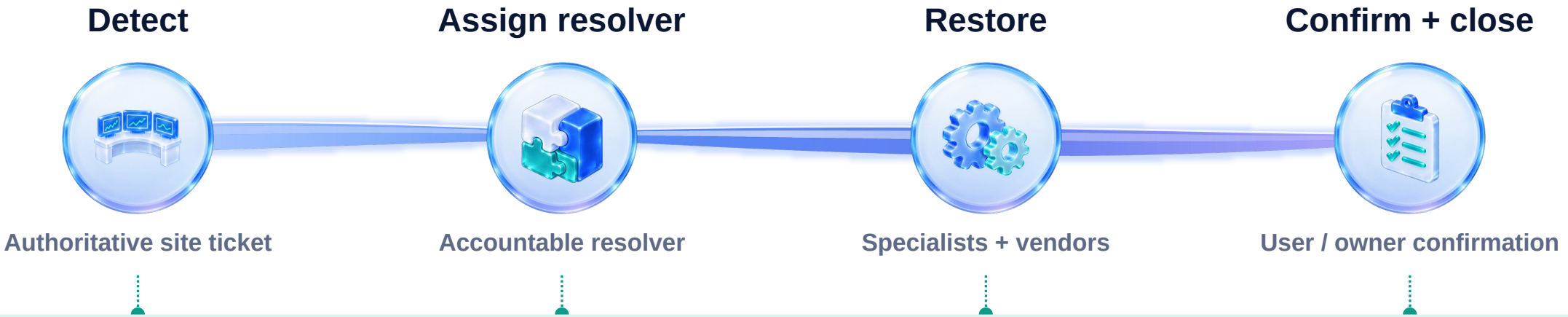
At day 120, compare aging, reassignment and handoff rework. NASA decides whether to expand, revise or stop.

RFP L.8 B3 / Pilot assumptions: accessible records, agreed baselines, available staff and authorized sharing.

- We test cross-site improvements through bounded handoffs and let measured service benefit guide NASA’s decision.

One incident. One communication owner

Technical resolution advances while the service desk retains communications.



Service desk

Retains communications ownership

Site Lead appoints an incident lead when multiple groups are involved.

NAS control room: 24/7. NCCS help desk: 08:00–18:00 ET, Monday–Friday.

- One incident lead coordinates recovery while one service desk keeps the researcher informed through closure.

Security across every service layer

Government officials accept risk

Govern	Approved plans and authority
Protect	Least privilege and configuration
Detect + respond	Findings and incident coordination
Recover	Verified restoration and closure

Productive scientific access

Weekly credentialed assessments

Annual audits of all resources

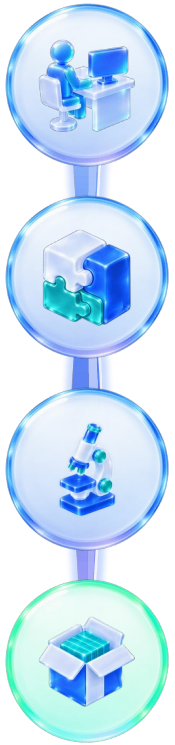
- Approved access, continuing assessment and verified recovery protect scientific work under NASA’s risk authority.

Procurement for useful capability

Alternatives compete on mission fit, test evidence and full implementation cost.

Delivered value

Full implementation cost



01 Mission need

Workload and acceptance criteria

02 Alternatives

Useful capability; install, train, operate and retire

03 Testbeds

Representative workload evidence

04 Integration

Shipment, receipt, acceptance and readiness

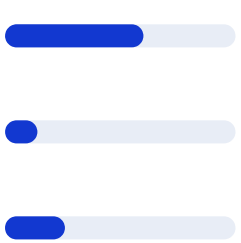
ODCs require prior written CO and COR approval.

- A purchase delivers value when mission fit, test evidence and the full cost of integration support the decision.

Procurement profile in the supplied RFP

RFP L.8 B4, Table 1 and Table 2, p. L-13

First-year estimate	NAS	NCCS	Combined
Compute expansion	$24m	$6m	$30m
Storage expansion	$6m	$1m	$7m
Infrastructure / maintenance	$10m	$3m	$13m
Total	**$40m**	**$10m**	**$50m**



250 purchases / $50m

Historical first-year planning profile: 150 NAS purchases and 100 NCCS purchases. The RFP says approved budgets may adjust these amounts.

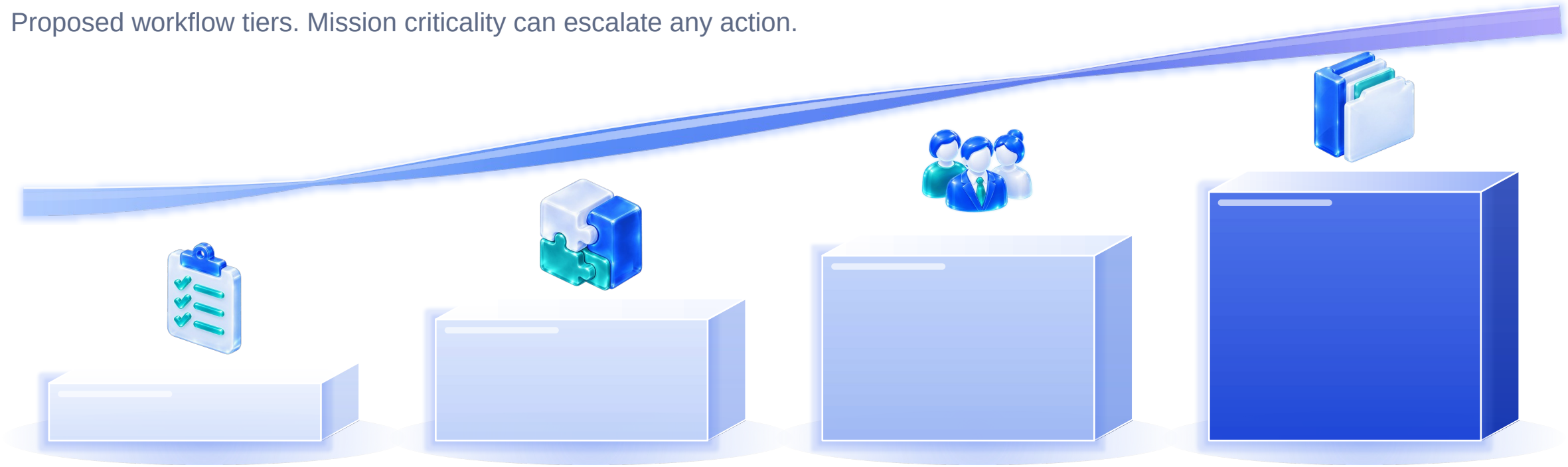
Delivery implication

Large compute / storage awards need coordinated integration. Numerous smaller purchases need efficient controls and maintenance continuity.

- The historical purchase profile calls for disciplined handling of both major systems and numerous smaller actions.

Review depth follows mission risk

Proposed workflow tiers. Mission criticality can escalate any action.



Below $10k

Compatibility and approval

$10k to below $100k

Alternatives and integration

$100k to $1m

Cross-functional risk review

Above $1m

Dedicated integration and acceptance plans

These bands describe the proposed review workflow and are not legal thresholds.

- Review effort grows with integration complexity and mission consequence; critical risk can escalate any action.

A visible path to commitment closure

NASA can inspect linked records and unresolved exceptions before acceptance.

Monthly Government reviews / Quarterly process reviews

Authorized scope	Owner + milestone	Test + configuration	Independent review	Acceptance
CO-issued work	Accountable lead	Recorded evidence	Quality check	Government decision

Exception: risk owner and recheck

- Every commitment closes through a named owner, inspectable evidence, independent review and NASA acceptance.

SECTION 05

Past performance

Relevant experience supported by attributable results.

Evidence behind the claim

Relevant scope

Comparable work

Verified results

Baseline + outcome

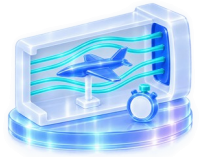
Client evidence

Traceable records

- Relevant scope, verified outcomes and attributable client evidence make a past-performance claim assessable.

Relevant experience and results

Fictional engagements and metrics created for this portfolio

	Sample engagement	Comparable scope	Illustrative result	NASA relevance
	Aster Research Consortium 2021–2024	HPC operations 2 sites / 46 staff $28.4m sample value	**28-day transition** 52 procedures with qualified alternates	Transition readiness Local ownership Continuity
	**Helios Aerospace Institute** 2022–2025	Scientific applications 18 staff $12.6m sample value	**38% shorter runtime** 20 science validation cases passed	Application scaling Scientific validity Usable handoff
	Peregrine Earth Systems Lab 2023–2025	Climate data services 16 staff $9.8m sample value	**59% less retrieval time** 120 datasets with traceable provenance	Researcher support Data services Reproducibility

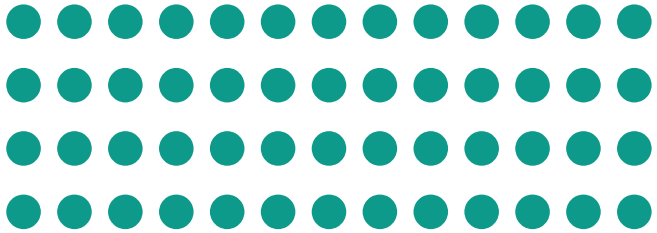
- Three fictional engagements illustrate how comparable scope and measurable outcomes support the delivery story.

FICTIONAL PAST PERFORMANCE

Aster Research Consortium

2021–2024 / HPC operations / 46 staff / 2 sites

52/52



> **“We knew who owned every handoff before day one.”**
>
> Fictional research operations director

Operating baseline

Active research and open work could not pause during a service-team change. Maya and Daniel owned transition readiness.

Demonstrated handover

Reconciled tasks and access, transferred runbooks and rehearsed recovery. Every transfer required a receiving owner.

28-day transition

52 of 52 critical procedures had qualified alternates before transfer. Evidence: readiness checklist, demonstrations and access register.

- The Aster example shows how demonstrated handovers and qualified alternates protect research during transition.

FICTIONAL PAST PERFORMANCE

Helios Aerospace Institute

2022–2025 / Aerodynamic simulation / 18 staff

RUNTIME

BASELINE

18.0 h

RESULT

11.2 h

> **“We gained time without losing confidence in the model.”**
>
> Fictional simulation program lead

Baseline: 18.0 hours

The 100-case ensemble could not finish overnight. Priya led diagnosis with investigators and system specialists.

Targeted engineering

Profiled communication and I/O, changed data layout and repeated timings using the same cases and allocation.

38% shorter runtime

Runtime fell to 11.2 hours. All 20 agreed science cases passed. The 6.8-hour reduction is 37.8%, rounded to 38%.

- The Helios example pairs shorter runtime with unchanged science checks to make the improvement credible.

FICTIONAL PAST PERFORMANCE

Peregrine Earth Systems Lab

2023–2025 / Climate data services / 16 staff

MEDIAN RETRIEVAL

BASELINE

6.4 h

RESULT

2.6 h

> **“Our researchers could find the right version and get to work.”**
>
> Fictional data services director

Baseline: 6.4-hour median

Researchers waited for large transfers and struggled to confirm dataset versions. Jordan owned service improvement.

Traceable distribution

Added server-side subsetting, versioned manifests and a named support owner for 120 curated datasets.

59% less retrieval time

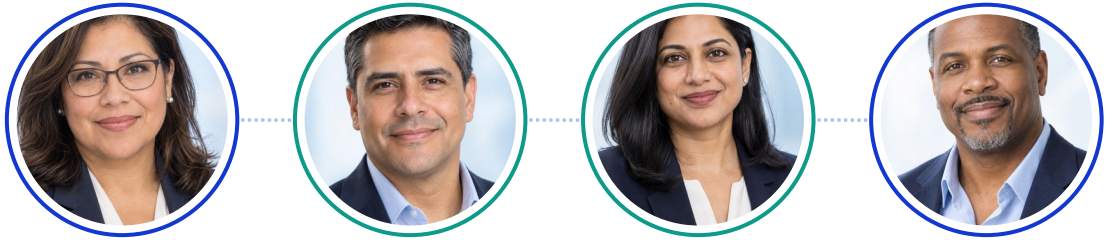
Median retrieval fell to 2.6 hours on the same 40-request set. The 3.8-hour reduction is 59.4%, rounded to 59%.

- The Peregrine example connects traceable data services to less waiting and clearer ownership for researchers.

SECTION 06

Key personnel

The people behind each commitment and result.



- The proposed key people connect personal experience and full commitment to the responsibilities NASA will rely on.

Maya Bennett

RFP L.8 A3 and M-7 / fictional resume

Program Manager

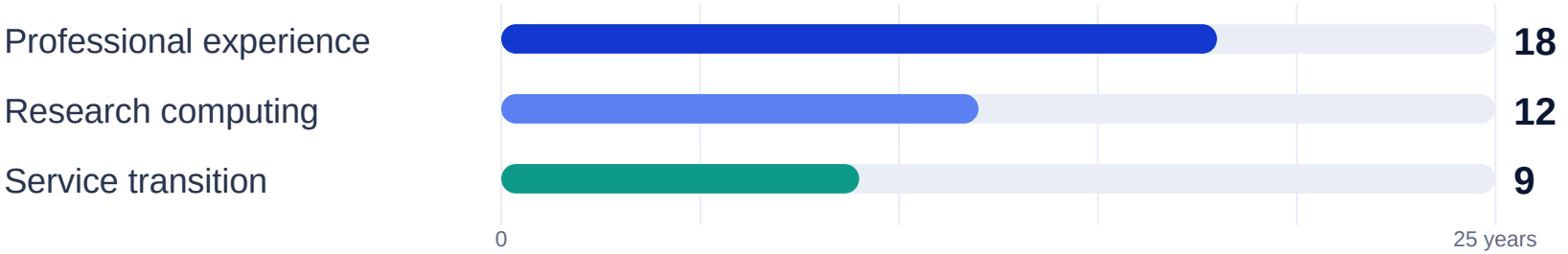
Mountain View, CA

100% proposed commitment

Fictional person / synthetic portrait

Leads complex research-computing services with responsibility for contract performance, staffing, customer decisions and transition readiness.

EXPERIENCE IN YEARS



Education and professional practice

M.S., Systems Engineering, Northbridge Institute of Technology, 2008

B.S., Engineering Management, Lakeshore University, 2006

PMP certification, 2018

Why this position is key

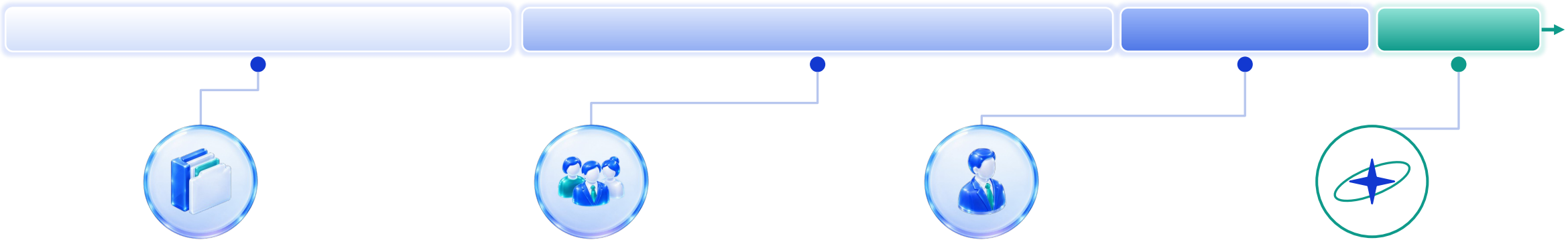
100% commitment is modeled because two-site performance, staffing decisions and NASA interfaces require continuous leadership. All planned time supports this contract.

- Maya’s proposed role brings program governance, research-service experience and transition ownership together.

KEY PERSONNEL

Maya Bennett / relevant experience

Program Manager / 100% proposed contract commitment



2008–2014

Lydian Computing Services

Systems Analyst

Developed service requirements, operational procedures and work-product review records.

2014–2021

Lydian Computing Services

Delivery Manager

Managed multi-team application support, service reporting, staff development and planned operating changes.

2021–2024

Aster Research Consortium

Program Manager

Led 46 people across two research sites. Completed the sample transition in 28 days and qualified alternates for 52 critical procedures.

2024–present

AXIS NOVA

Program Director

Leads the fictional two-site delivery model and coordinates corporate staffing, contracts and quality support.

- Maya’s fictional career connects hands-on delivery experience to responsibility for the full research service.

Daniel Reyes

RFP L.8 A3 and M-7 / fictional resume

NAS Site Delivery Lead

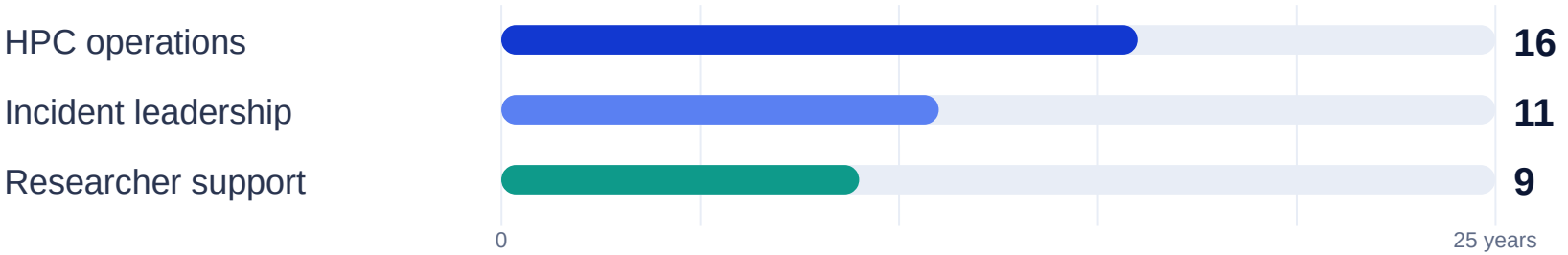
Mountain View, CA

100% proposed commitment

Fictional person / synthetic portrait

Directs local HPC service delivery with experience in control-room coverage, incident coordination, operational qualification and researcher support.

EXPERIENCE IN YEARS



Education and professional practice

B.S., Computer Engineering, Pacific Technical University, 2010

ITIL 4 Foundation, 2020

Linux and HPC operations practice

Why this position is key

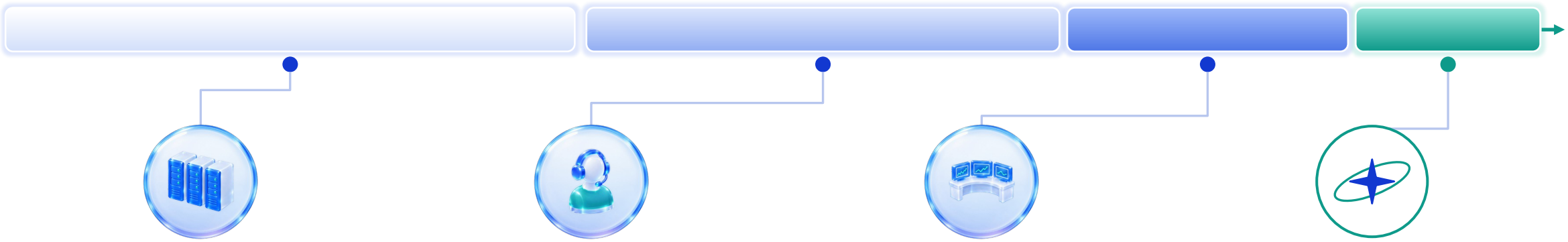
100% commitment supports continuous local management of the NAS service. Shift personnel provide coverage across 24 hours. The key-person allocation does not claim that one manager staffs every shift.

- Daniel’s proposed role anchors local NAS operations, incident leadership and qualified round-the-clock coverage.

KEY PERSONNEL

Daniel Reyes / relevant experience

NAS Site Delivery Lead / 100% proposed contract commitment



2010–2016

Lydian Computing Services

HPC Engineer

Supported Linux clusters, schedulers, software changes and operational documentation.

2016–2021

Lydian Computing Services

Incident Manager

Coordinated storage and compute incidents, recovery exercises, vendor escalations and user communications.

2021–2024

Aster Research Consortium

Operations Lead

Qualified alternates for 52 critical procedures. Reduced sample ticket reassignments from 25% to 11% through resolver ownership and runbook changes.

2024–present

AXIS NOVA

NAS Delivery Lead

Maintains the sample NAS operating model, shift-readiness reviews and interfaces with application and facility specialists.

- Daniel’s fictional career shows increasing responsibility for HPC operations, incidents and dependable site delivery.

Priya Shah

RFP L.8 A3 and M-7 / fictional resume

Technical Integration Lead

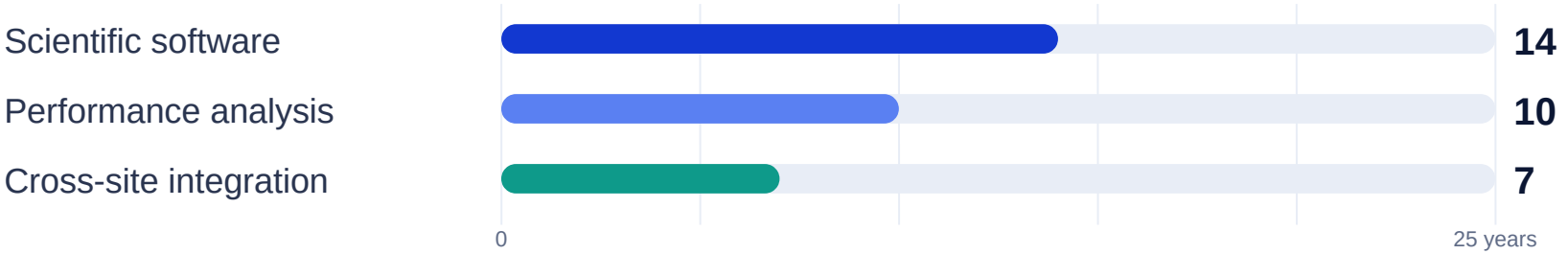
Mountain View, CA

100% proposed commitment

Fictional person / synthetic portrait

Combines scientific application engineering with cross-site integration, benchmark design and evidence-based technology evaluation for research users.

EXPERIENCE IN YEARS



Education and professional practice

Ph.D., Computational Science, Westhaven Institute of Technology, 2012

B.S., Applied Mathematics, Lakeshore University, 2007

MPI / OpenMP and parallel-performance practice

Why this position is key

100% commitment reflects the modeled demand for simultaneous application diagnostics and integration across NAS and NCCS. NASA-approved tasks determine the actual assignment of effort.

- Priya’s proposed role connects scientific application engineering to cross-site dependencies and evidence for change.

KEY PERSONNEL

Priya Shah / relevant experience

Technical Integration Lead / 100% proposed contract commitment

2012–2017

Meridian Simulation Works

Computational Scientist

Developed parallel research codes, numerical test cases and scientific workflow documentation.

2017–2022

Meridian Simulation Works

Senior Scientist

Profiled distributed applications, evaluated memory and I/O limits, and developed reproducible validation and release evidence.

2022–2025

Helios Aerospace Institute

Application Lead

Reduced the sample 100-case ensemble runtime from 18.0 to 11.2 hours on the same allocation. All 20 agreed science-validation cases passed.

2025–present

AXIS NOVA

Technical Integration Lead

Leads the sample exascale diagnostic, technology roadmap and cross-site service-interface reviews.

- Priya’s fictional career connects computational science, performance diagnosis and accountable technical integration.

Jordan Ellis

RFP L.8 A3 and M-7 / fictional resume

NCCS Site Delivery Lead

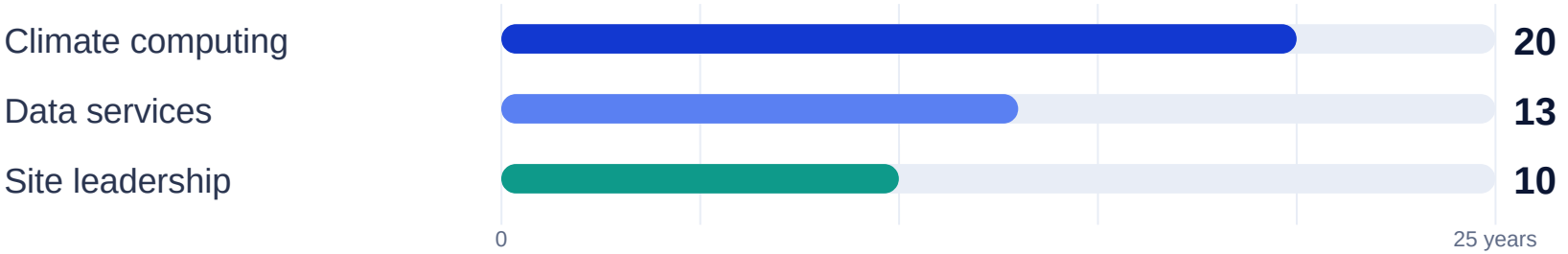
Greenbelt, MD

100% proposed commitment

Fictional person / synthetic portrait

Leads climate-computing and research-data services with experience in dataset access, provenance, scientific-user support and site operations.

EXPERIENCE IN YEARS



Education and professional practice

M.S., Applied Mathematics, Atlantic Research University, 2006

B.S., Computer Science, Northbridge Institute of Technology, 2003

ITIL 4 Foundation, 2020

Why this position is key

100% commitment supports the modeled local management and research-data workload. Operational staff provide the required help-desk coverage while Jordan owns management and escalation.

- Jordan’s proposed role combines climate-computing knowledge, data-service ownership and local NCCS leadership.

KEY PERSONNEL

Jordan Ellis / relevant experience

NCCS Site Delivery Lead / 100% proposed contract commitment

2006–2016

Meridian Climate Services

Scientific Programmer

Built climate-data processing workflows, metadata checks and analysis tools for distributed researchers.

2016–2023

Meridian Climate Services

Site Manager

Managed research support, data-service changes, recovery planning and staffing readiness for climate analysts.

2023–2025

Peregrine Earth Systems Lab

Data Services Lead

Improved access to 120 curated datasets. Reduced median retrieval from 6.4 to 2.6 hours for the same 40-request sample evaluation set.

2025–present

AXIS NOVA

NCCS Delivery Lead

Owns the sample NCCS local delivery plan and interfaces among user services, climate applications and data distribution.

- Jordan’s fictional career connects climate workflows and traceable data access to responsibility for site service.

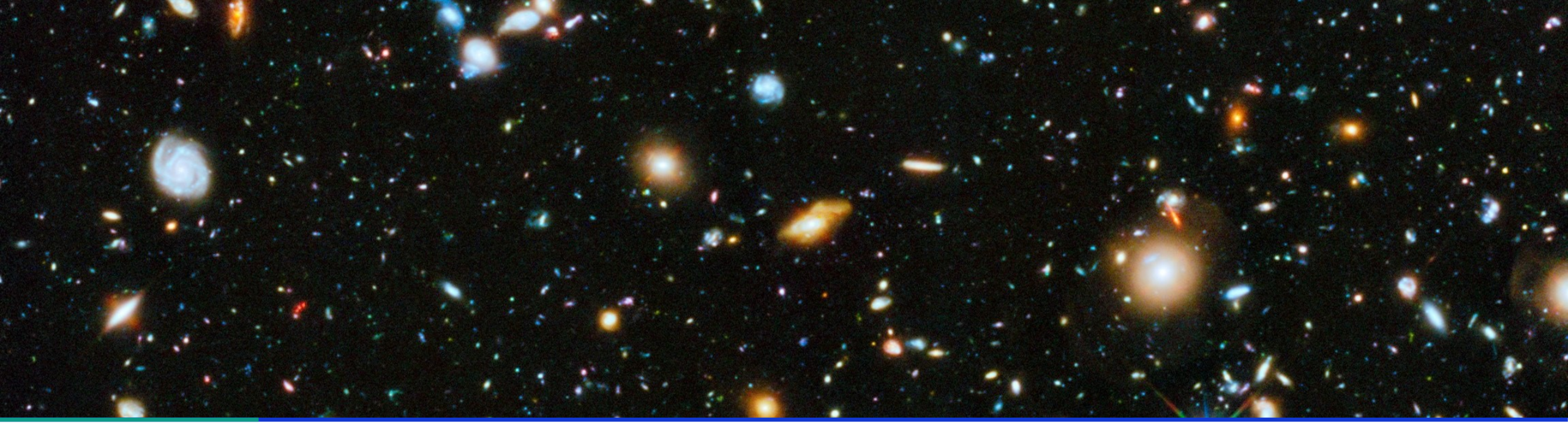
Key-personnel authority, commitment and replacement

RFP L.8 A3; M-7; H.2 / fictional proposed assignments

Key position / prime-held	Authority and principal assignments	100% proposed time allocation
Maya Bennett Program Manager	Directs total contractor performance, resources and priorities. Escalates contractual scope, cost and obligation changes to the CO.	40% Program governance 25% Staffing + resources 20% Quality + risks 15% NASA coordination
Daniel Reyes NAS Site Delivery Lead	Directs approved NAS work and local staff. Assigns incident leadership; escalates shared-resource conflicts, security concerns and changes outside authority.	45% Local delivery 25% Workforce readiness 20% Incidents + changes 10% NASA reporting
Priya Shah Technical Integration Lead	Coordinates cross-site technical dependencies and recommends options. Site leads retain operations; NASA approves work and investigators define science criteria.	35% Technical integration 30% Application assurance 20% Roadmap + pilots 15% Science coordination
Jordan Ellis NCCS Site Delivery Lead	Directs NCCS staff and approved site work. Coordinates data-service and support incidents; escalates scope, security and resource decisions.	40% Local delivery 30% Data-service coordination 20% Workforce readiness 10% NASA reporting

Key-person changes require CO consent under H.2. Clarification 02 explains coverage and replacement timelines.

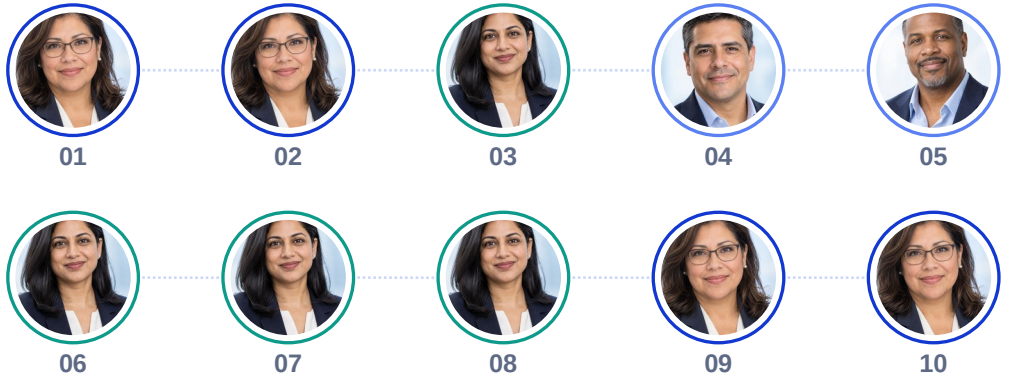
- Defined authority, explicit time allocation and controlled replacement make the proposed key-person commitments visible.



SECTION 07

Government questions

Ten likely evaluation questions
with clear, accountable responses.

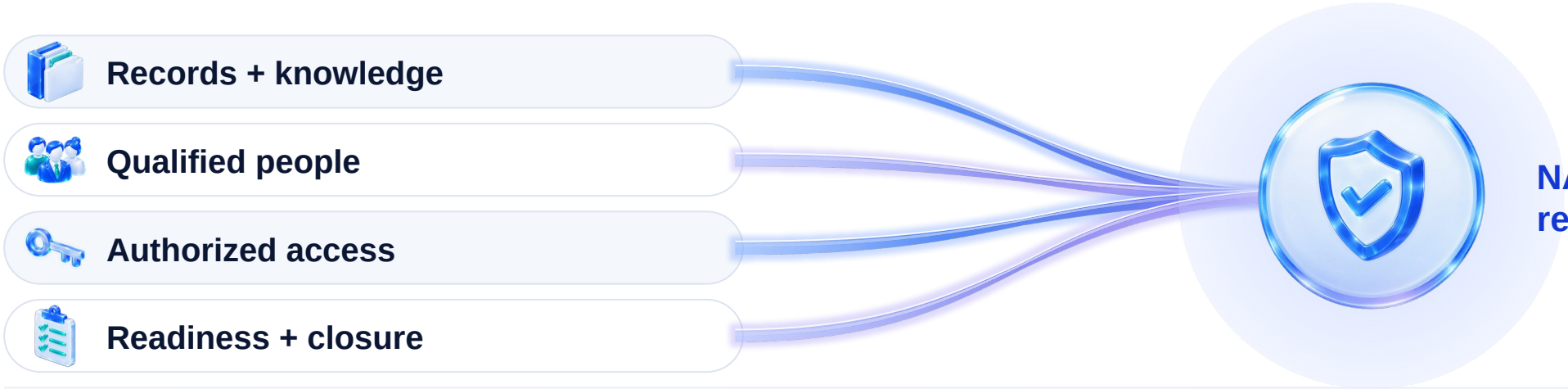


- These illustrative Government questions test how our approach holds up when delivery becomes difficult.

Maya Bennett / Program Manager

How will you protect NASA service during phase-in?

Responsibility transfers only after qualified people, access and operating procedures are ready.



Maya reconciles active tasks, property, incidents and coverage in days 1–5 with the outgoing team and NASA.

Site leads transfer runbooks and recovery knowledge, then demonstrate procedures without disrupting ongoing research.

NASA reviews readiness and blockers before transfer. A missing qualification or access permission triggers qualified recovery and a revised sequence.

EVIDENCE Readiness roster, task / property reconciliation, access status, procedure demonstrations and signed review record.

- We protect research during phase-in by proving readiness before NASA agrees to the transfer of responsibility.

Maya Bennett / Program Manager

What happens if a key person leaves or cannot start?

Maya protects critical coverage immediately and submits a controlled replacement for CO consent.

Immediate coverage

Notify NASA promptly, identify affected decisions and activate a qualified acting lead within approved authority.

Candidate review

Review candidates by ten business days and target a complete substitution package by twenty business days.

Documented handover

Target a start within thirty calendar days, subject to consent, notice and access. If a dependency slips, update the coverage and recovery plan.

CO consent

EVIDENCE Coverage matrix, proposed substitute resume, commitment allocation, impact assessment and handover record.

- Immediate qualified coverage protects the work while a documented replacement follows the required consent process.

Priya Shah / Technical Integration Lead

Who is accountable when work crosses Ames and Goddard?

One task lead owns the work. Local site leads retain operations, and Priya coordinates the shared dependency.

NAS / Daniel

NAS Pleiades storage · NASA Ames/Dominic Hart

Shared technical integration

NCCS / Jordan

Discover supercomputer · NASA Goddard

The approved task names the technical lead, Government interface, delivering team and receiving owner.

Priya records interface requirements and handoff criteria. Daniel and Jordan verify local readiness and protect site coverage.

Maya resolves contractor resource conflicts. Any change to contractual obligations returns through the CO process.

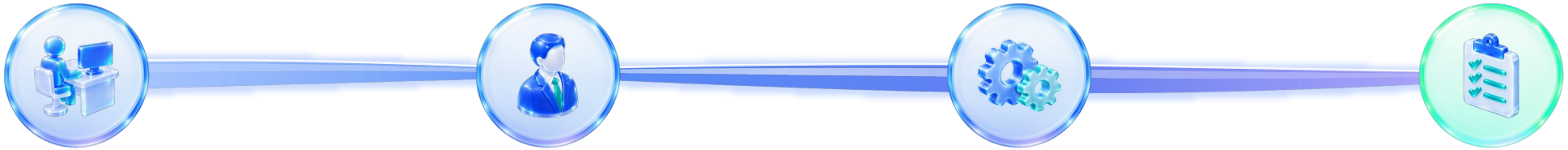
EVIDENCE Task record, responsibility matrix, interface register, handoff acceptance and decision history.

- Cross-center work has one task owner, local operating leads and a named coordinator for the shared dependency.

Daniel Reyes / NAS Site Delivery Lead

How will you manage an incident that stops research?

The site lead assigns a technical incident lead while the service desk retains user communications.



Research impact | **Assigned incident lead** | **Verified recovery** | **Documented closure**

One service desk owns user communications

Record the affected research, deadline, system symptoms and severity in one event log. Preserve evidence and notify the responsible NASA interface.

Coordinate compute, storage, network, application and vendor specialists. Use the approved recovery procedure and record decisions.

Confirm restored service with the affected owner, verify corrective actions and update the runbook before closure.

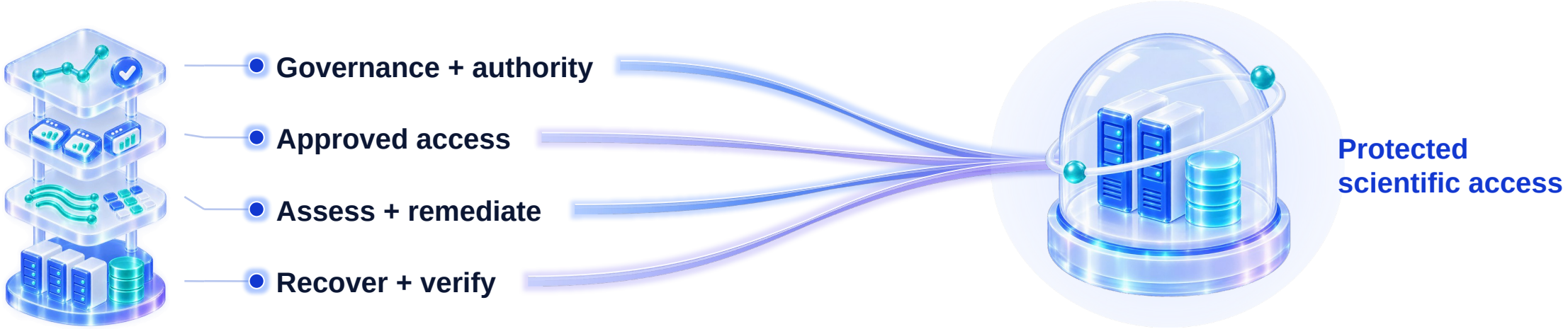
EVIDENCE | Event timeline, communications, recovery checks, user verification and corrective-action closure.

- Research impact drives incident leadership, coordinated recovery and verified closure with one communication owner.

Jordan Ellis / NCCS Site Delivery Lead

How will you secure access while supporting NASA science?

Access planning and continuous assurance follow NASA’s approved security authority and research needs.



Coordinate identity, training, allocation and least-privilege permissions before the researcher needs the service. Assess resources before network connection.

Run credentialed vulnerability assessments at least weekly, provide findings to administrators, and track remediation with owners and retests.

Support annual resource audits. Investigate indications and refer hostile acts to center security and NASA SOC. Government officials retain risk acceptance.

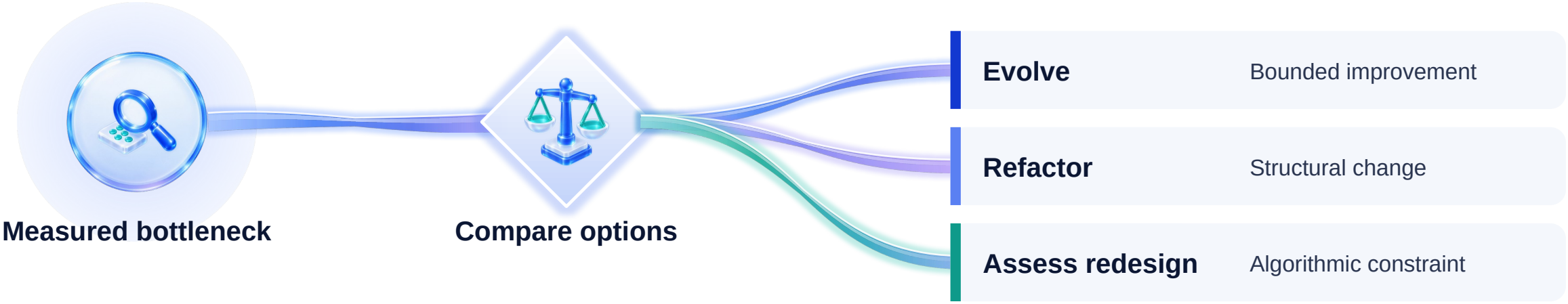
EVIDENCE Access approvals, assessment results, remediation records, retest evidence and escalation history.

- We sustain scientific access through approved permissions, recurring assurance and NASA’s acceptance of risk.

Priya Shah / Technical Integration Lead

How will you decide whether an application needs refactoring?

We compare measured scaling limits and engineering options while preserving the scientific reference.



Baseline representative cases, process / thread counts, libraries and allocation. Profile communication, memory, I/O, imbalance and recovery cost.

Test a bounded local improvement against a structural alternative when the evidence indicates decomposition or data layout limits.

Repeat uninstrumented timing and investigator-approved science checks. Recommend the least risky maintainable option that meets the research objective.

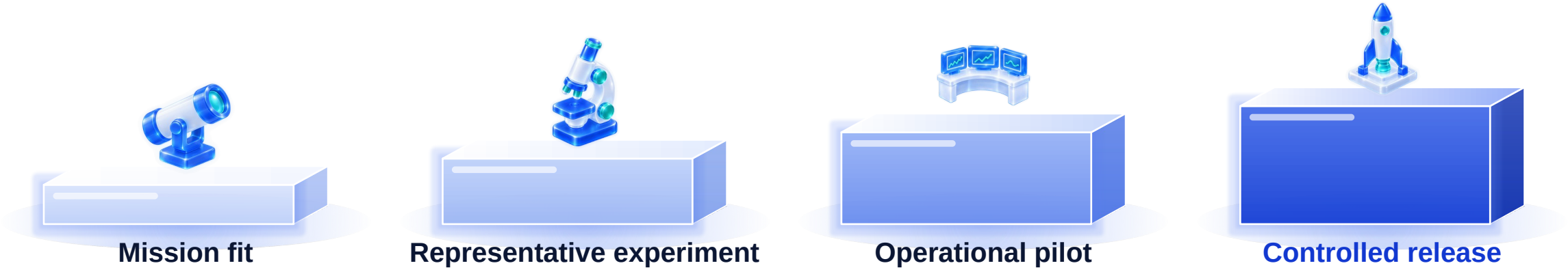
EVIDENCE Scaling plots, profile traces, code changes, timing repeats, science cases and the alternatives analysis.

- We compare measured constraints and practical alternatives before recommending a maintainable engineering change.

Priya Shah / Technical Integration Lead

When would you recommend an accelerator or AI method?

Only after NASA-relevant evidence supports a useful benefit and an operable, reversible introduction.



Mission fit

Representative experiment

Operational pilot

Controlled release

Define the mission problem and compare with the current method. Identify portability, data, security and skills dependencies.

Use representative benchmarks to assess the complete workflow. For AI-assisted analysis, test scientific error, limitations and drift against reference cases.

Advance through experiment and pilot gates only when evidence supports progression. Recommend a controlled release with training, support and fallback.

EVIDENCE | Benchmark method, comparison results, scientific limitations, risk record, readiness and rollback plan.

- An accelerator or AI method earns a place through NASA-relevant evidence and a reversible operational introduction.

Priya Shah / Technical Integration Lead

How will a procurement deliver useful capacity on schedule?

Workload acceptance and site readiness shape the purchase from the start.

Translate the NASA workload into requirements and tests. Compare reuse, expansion and alternative solutions with the affected site.

Align vendor delivery with facilities, networking, security, software and operator readiness. Track long-lead dependencies and maintenance continuity.

Obtain required approvals, verify receipt, test the installed capability and complete the Government acceptance process before production handoff.

EVIDENCE Requirements-to-test matrix, alternatives, approval record, delivery milestones and acceptance results.

- Workload acceptance, site readiness and controlled integration shape a procurement before production handoff.

Maya Bennett / Program Manager

How will you handle changing priorities or additional work?

We make the consequences visible and obtain authorization before changing contractual obligations.

Requested outcome → **Impact and options** → **CO authorization** → **Updated task baseline**

The task lead records the request, existing authority and impacts on science milestones, staffing, dependencies and budget.

Maya compares resequencing, deferring lower-priority work and adding qualified resources. Each option identifies affected commitments and a recovery owner.

The CO authorizes contractual changes. The team updates the baseline, communicates the decision and tracks revised commitments.

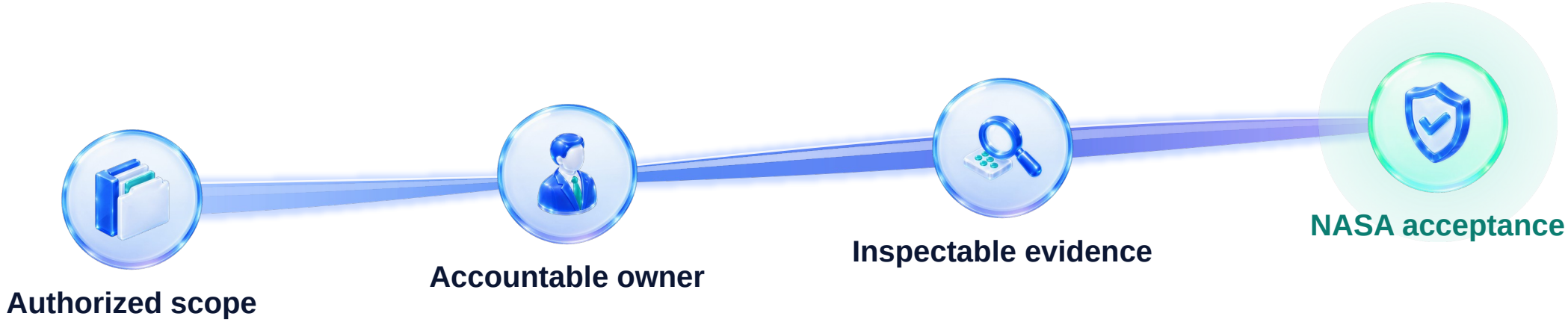
EVIDENCE | Impact assessment, options, decision log, CO authorization when required and updated task record.

- We make the consequences of changed priorities visible and obtain authorization before changing obligations.

Maya Bennett / Program Manager

Why should NASA have confidence in this delivery approach?

The approach connects required work to qualified people, measurable evidence and Government acceptance.



Named leaders own contractor decisions, local continuity and cross-site interfaces. Staffing assumptions and partner workshare remain visible.

Technical recommendations include science validation, assumptions and risk responses. Implementation plans identify milestones and deliverables.

NASA can inspect progress and unresolved exceptions throughout delivery. The fictional cases demonstrate how comparable evidence would support the selection narrative.

EVIDENCE Traceable task records, personnel qualifications, test results, risk register and verified corrective actions.

- NASA can judge delivery through qualified owners, inspectable results, visible exceptions and acceptance evidence.

Science comes first

- A qualified team on day one.
- A clear owner for every task.
- Evidence behind every change.

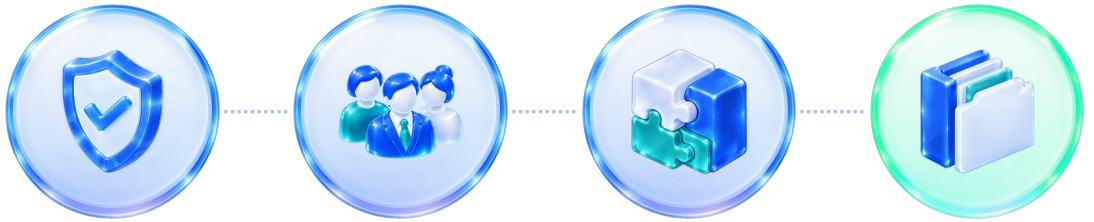
Ready to support the next discovery.

- Our commitment is a qualified team, an owner for every task and evidence behind every change in service of science.

APPENDIX

Supporting discussion

Written-topic concepts outside the core oral talk



- The supporting controls make the delivery commitments practical, from independent judgment to specialist participation.

Independent information and judgment

Three conflict mechanisms require different controls.

Unequal access

Need-to-know repositories
Access and distribution controls

Impaired objectivity

Independent reviewer
Recuse and replace

Biased ground rules

Restrictions and disclosure
Covered prime / first-tier awards

H.23 restrictions, H.4 protections and flowdowns, and written CO disclosure.

- Different conflict mechanisms receive specific controls so information access and technical judgment remain independent.

Workforce confidence and continuity

Incumbent treatment is verified individually. No automatic transfer is assumed.

Qualified coverage

Supervisor + HR review

Skill continuity

Market alignment

Role and compensation review

Internal equity

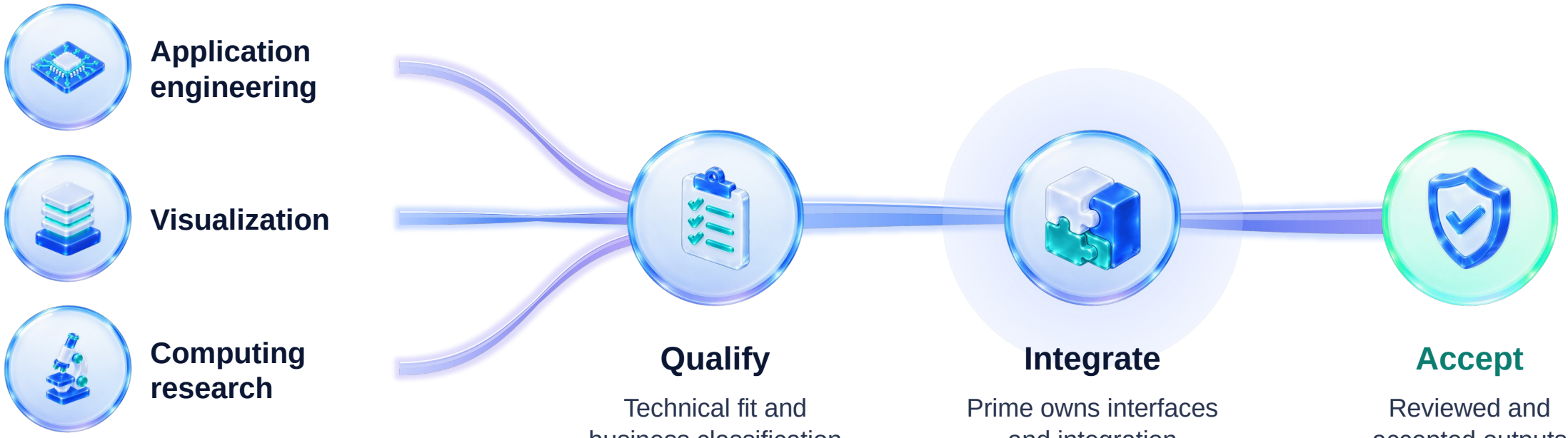
Consistent treatment

Employee package

Offer and verified classification

- Individual qualification and fair, verified employment practices support continuity without assuming automatic transfer.

Meaningful specialist participation



Sample partners: Orbital Method Labs / Lumen Scientific Studio / Vector Ridge Research.

- Specialists contribute through qualified work packages, managed interfaces and outputs the prime reviews and accepts.

Evaluation traceability / find the supporting evidence

Historical NACS RFP L.8 and Section M; J.1(a) SOW

RFP topic	Where the story answers it	Evidence NASA can inspect
A1 Management	Authority, task control and phase-in	Decision rights, task plan, readiness record
A2 Staffing	Capture basis, recruiting, retention, partners	Roster, qualification matrix, workshare ledger
A3 Key personnel	Profiles, career history, assignment matrix	Education, duties, authority, time commitment
B1 Technology roadmap	Stages, risks, gates and delivery timing	Workload baseline, pilot evidence, fallback
B2 Exascale	Diagnosis, engineering choices, validation	Science tests, repeatable timing, risk register
B3 Service optimization	Local operating model and common processes	Process milestones, handoffs, service measures
B4 Procurement	Alternatives, profile and review depth	Requirements, integration and acceptance tests
Past performance	Three fictional comparable engagements	Scope, ownership, measured result and method

A visual crosswalk supports evaluator navigation; the complete solicitation still governs every submission requirement.

- This crosswalk connects each evaluation topic to the delivery story and the evidence NASA can inspect.

APPENDIX

Portfolio basis and source notes

Historical NASA requirements, illustrative responses and clearly identified portfolio fiction.

Requirements basis

2017 NACS RFP NNA17554082R and J.1(a) SOW. Management, staffing, scientific applications, operations and procurement inform the story.

Fictional portfolio content

Company, people, credentials, partners, project histories, metrics and quotes are invented. Portraits and facility art are synthetic.

Format and boundaries

The 16:9 visual portfolio is not a formal submission. Ten Q&A prompts are illustrative. Written resumes and commitment documents remain separate.

IMAGE CREDITS

Photography: NASA Image and Video Library (images.nasa.gov), public domain; individual credits appear on each slide. Images are illustrative and do not imply NASA endorsement.

No NASA award or endorsement is represented. Text, tables and labels are editable. Dimensional illustrations are image-based.

- Historical requirements frame the example; all people, companies, projects, metrics and customer quotes are fictional.